(1) 13. Use an understanding of the context to guide the mathematical solving process

The question involves a **financial model for a wind power station**, requiring students to establish a mathematical model within a real-world context (construction cost, revenue from electricity generation, maintenance costs) and solve for the "break-even period."

- It requires understanding what "covering the cost" means, i.e., total revenue ≥ total investment.
- Students must make reasonable assumptions and calculations based on the provided formula and parameters (e.g., operational days per year = 97% × 365) and choose a **realistically appropriate level of accuracy**.

(2) 12. Use a multi-step procedure leading to a mathematical solution

Solving requires following a **multi-step process**:

1. Understand the meaning of each term in the formula (e.g., annual revenue, annual cost).
2. Substitute given values (unit price, efficiency, construction cost, etc.).
3. Set up the inequality "total revenue ≥ construction cost" and solve for the number of years y.
4. Compare the options and select the **smallest integer year** that satisfies the condition.

(3) 7. Manipulate numbers, algebraic expressions, and equations

- Involves **numerical calculations** (e.g., unit conversion, percentage operations).
- Requires solving an **algebraic inequality** for the variable y.
- While no graph is directly provided, the formula itself is a form of **mathematical representation**.

(4) 8. Articulate a solution, showing intermediate mathematical results

Students are expected to **demonstrate calculation steps**, such as:

- Calculating annual electricity generation
- Deriving the annual net revenue formula
- Showing intermediate steps in solving the inequality
  Finally, they must **clearly state the "minimum number of years" and match it with the correct option**.

(5) 3. Implement a given strategy to determine a mathematical solution

The problem provides a **financial gain formula**, which is a "given strategy."
Students must correctly **apply this formula**, set up the inequality in terms of years, and solve it.

(6) 10. Use a variety of representations when interacting with a problem

- Textual description (problem context)
- Mathematical formula (the mayor's given formula)
- Table of numerical parameters (model specifications)
- List of options (multiple choice)
  Students need to translate and integrate information across these different forms.

（1）13. 利用情境理解引导数学解题过程

题目涉及风电站的**财务模型**，要求学生在实际情境（建设成本、发电收益、维护费用）中建立数学模型并求解“回本年限”。

- 需要理解“成本覆盖”的含义，即总收入 ≥ 总投资。
- 需要根据提供的公式和参数（如年运行天数 = 97% × 365）进行合理假设和计算，选择**现实合理的精度**。

（2）12. 使用多步程序得出数学解、结论或概括

解题需遵循**多步流程**：

1. 理解公式中各项的含义（如年收益、年成本）。
2. 代入已知数值（单价、效率、建设成本等）。
3. 建立不等式“总收益 ≥ 建设成本”并解出年份 y。
4. 比较选项，选择满足条件的**最小整数年**。

（3）7. 操作数字、图表和统计信息、代数表达式和几何表示

- 涉及**数值计算**（如单位换算、百分数运算）。
- 需要处理**代数不等式**并解出变量 y。
- 虽未直接给出图表，但公式本身是一种**数学表征形式**。

（4）8. 清晰表达解题过程，展示或总结中间数学结果

要求学生能**展示计算步骤**，例如：

- 年发电量计算
- 年净收益公式推导
- 解不等式的中间步骤
  最终需**明确表达“最小年数”并匹配选项**。

（5）3. 使用给定策略确定数学解

题目已给出**财务增益公式**，属于“给定策略”。
学生需正确**应用该公式**，建立关于年份的不等式并求解。

（6）10. 在问题解决中使用多种表征形式

- 文字描述（问题背景）
- 数学公式（市长给出的公式）
- 数值参数表（模型规格）
- 选项列表（多项选择）
  学生需在不同形式之间转换理解，整合信息。